**Sankalp Jha**
Open Source Developer | Software Engineer
Delhi NCR, India

**Portfolio** | **LinkedIn** | **GitHub**
sankalp.jha9643@gmail.com
**+91-8800117422**

## Education

| Year | Degree/Certificate | Institute | CPI/% |
|---|---|---|---|
| **Oct 2022 – Aug 2026** | B.Tech in CS & IT | Ajay Kumar Garg Engineering College, Delhi NCR | **7.25/10** |
| **2021 – 2022** | Senior Secondary (XII) | New Era School, Delhi NCR | **91.2%** |
| **2017 – 2020** | High School (X) | SJN International School, Coimbatore | **92.6%** |

## Experience

**Google Summer of Code 2025 Contributor** | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)
**SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4** *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- **Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by **30%**, and optimized SID recirculation to achieve **10** Gbps line-rate throughput with **20%** lower inference latency.

## Open Source Contributions

- **WasmEdge/WasmEdge:**
  **#4470** fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch
  **#4440** Upgrade Alpine base to **3.23** and restore static LLD support
  **#4466** feat(docker): add standalone alpine-base image build pipeline
  **#4452** docs(README): fix broken introduction and blog links
- **Project-HAMi/HAMi:**
  **#2292** Corrected hardware allocation logic for Kunlun and AWS Neuron AI accelerators to prevent invalid state generation
  **#2307** Eliminated node-level deadlocks by strictly enforcing lock releases during failed device-binding operations
  **#2259** Stabilized cluster scheduling by adding safe state handling for deleted ResourceQuota tombstones
  **#2204** Prevented metrics panics by guarding hardware memory ratio calculations against unknown total memory states
  **#2221** fix(workflow): skip non-translatable issue comments
- **facebook/bpfilter:**
  **#507** build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c
- **p4lang/tutorials:**
  **#681** Update README.md in reference to #**680**

## Key Projects

**Cutable** | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (*Project Link*)

**Objective:** Generate editable React applications from text and visual requirements.
**Approach:** Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.
**Impact:** Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

**DMIE** | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (*Project Link*)

**Objective:** Determine whether a startup idea already exists by comparing it against more than **500,000** company records.
**Approach:** Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.
**Impact:** Raised USD **10,000** and serves more than **100** users worldwide.

**Slix** | *React, Go, Azure OpenAI, REST API* (*Project Link*)

**Objective:** Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.
**Approach:** Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.
**Impact:** Gives viewers a concise representation of a movie's tone before watching.

**Myprod** | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (*Project Link*)

**Objective:** Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.
**Approach:** Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.
**Impact:** Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

**XNIC v1** | *Linux Kernel, C, PCI, BAR0 MMIO, DMA, Descriptor Rings, e1000, Intel 82540EM, net_device, NAPI, IRQ, MSI, ethtool, QEMU, HVF, KFENCE, sparse, DPDK, rte_ethdev, net_pcap PMD, tcpdump, Wireshark, W5500, SPI, Device Tree, Bash, GitHub Actions, Vercel* (*Project Link*)

**Objective:** Build and qualify a defensible clean-room Linux Ethernet driver with observable packet paths, ownership invariants, fault recovery, and a gated user-space DPDK forwarding path.
**Approach:** Implemented PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX queue backpressure, ethtool counters, and serialized watchdog/reset recovery in C; added staged fault injection, KFENCE and sparse qualification, reproducible QEMU/HVF automation, Wireshark-compatible captures, a W5500 SPI driver path, and an rte_ethdev L2 forwarder.
**Impact:** Qualified **10,100** RX and TX ring wraps across **646,400** packets at zero loss, **1,000** interface cycles, **100** driver rebinds, and **100** reset-under-traffic cycles; validated DPDK partial-TX cleanup and signal-safe shutdown with the net_pcap virtual PMD.

## Technical Skills

- **Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- **Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- **Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- **Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

## Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
  - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a **60%** bump in engagement.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  - Co-led **10+** tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the **31+** crew members.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

## Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026*** *International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. **2026**. IEEE Xplore ↗ DOI: 10.1109/ICCSC67078.2026.11468597

## Certifications & Honors

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|---|---|
| **Certifications** | • **Open Science 101** – NASA • **Data Science for Engineers** – NPTEL • **Enhancing Soft Skills & Personality** – NPTEL • **IIRS Satellite Remote Sensing** – ISRO |
| **Honors & Awards** | • **2x** AKTU State Men Singles TT Gold (**2023**, **2024**) • International Robotics Olympiad Bronze, Thailand (**2019**) • Merit Certification at Carnic Model United Nation (**2021**) • NCC Cadet, National Cadet Corps (**2019**) |